

# High-Performance Polyurea Improved by Reactive Nanocluster for Antibiofouling

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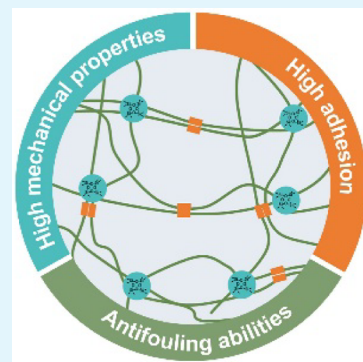
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Supporting Information

**ABSTRACT:** Polyurea has found applications in protective coatings. Yet, the too fast polymerization and lack of functions limit its application. Herein, we report a high-performance polyurea via the stepwise polymerization of an isocyanate (NCO)-terminated prepolymer consisting of poly(propylene glycol)-*block*-poly(ethylene glycol)-*block*-poly(propylene glycol) (PPG-*b*-PEG-*b*-PPG) with a nanocluster synthesized via the hydrolysis of *N*-phenylaminomethyltriethoxysilane. Such a nanocluster contains low-reactivity secondary amines, so the polymerization of polyurea can be slowed down (over 1 h), which improves its wetting and adhesion to a substrate. The residual silanol groups on the nanocluster further increase the adhesion. Such polyurea exhibits high adhesion on various substrates, including glass, ceramic, steel, copper, titanium, wood, and natural rubber ( $\sim 2.35$ – $14.64$  MPa). Besides, the nanoclusters can cross-link the prepolymer into a tough network, endowing the polyurea with a high mechanical strength of  $\sim 25$  MPa, much higher than the traditional polyaspartic ester polyurea. On the other hand, the PEG segments enable the polyurea to have good fouling resistance against proteins (fibrinogen absorption was reduced by over 90%), bacteria (RBA of *S. aureus*, *E. coli* and *Pseudomonas* sp. was less than 10%), as well as diatom (diatom density was less than 100 cells/mm<sup>2</sup>). The polyurea is expected to find applications in biomedical engineering and marine antifouling.

**KEYWORDS:** polyurea, nanocluster, antifouling, protein resistance, high adhesion



## INTRODUCTION

Biofouling formed by the attachment and accumulation of organisms is of great concern in a wide range of fields, such as biomedical,<sup>1,2</sup> biosensors,<sup>3</sup> textile and garment,<sup>4</sup> food packaging,<sup>5,6</sup> sewage disposal,<sup>7</sup> and marine industry,<sup>8</sup> seriously threatening human body, and bringing huge economic losses. To solve the problem, coating,<sup>9</sup> mechanical cleaning,<sup>10</sup> electrochemical treatment,<sup>11</sup> and ultrasonic cleaning<sup>12</sup> have been used. Actually, antifouling coating is the most used one because it is the most convenient, efficient, and economical. However, coatings containing biocides usually cause irreversible damage to the human body and the ecological environment.<sup>13</sup> PEG-based polymers with good protein resistance are considered to be used in antifouling field.<sup>14–19</sup> Yet, they often suffer from poor mechanical properties.<sup>20</sup> Besides, their hydrophilicity enables them to swell and peel from the substrate when immersed in water.<sup>21–24</sup>

Polyurea with excellent mechanical properties has found application in protective coatings.<sup>25–28</sup> In particular, polyurea has received more attention in the antifouling fields. For example, polydimethylsiloxane (PDMS)-based fouling-release polyurea has been prepared.<sup>29</sup> Attributed to the low surface energy and low elastic modulus of PDMS, fouling organisms cannot adhere firmly to its surface. However, since it does not contain antifouling groups, its static antifouling ability is poor. Moreover, the linear PDMS chain is soft, resulting in a low

mechanical strength of the polyurea. To improve the static antifouling ability, an antifoulant (4,5-dichloro-2-n-octyl-4-isothiazolin-3-one) was introduced into the PDMS-based polyurea by physical blending.<sup>30</sup> However, antifoulants have cytotoxic effects and may affect the marine ecosystem. In addition, its mechanical properties are still weak. To improve the mechanical properties, nanodiamonds were introduced to reinforce the PDMS-based polyurea.<sup>31</sup> However, the non-reactive nanodiamond may agglomerate during the curing process and significantly affect the polyurea performance. Moreover, due to the lack of antifouling groups, the static antifouling ability of nanodiamond reinforced PDMS-based polyurea is still poor. In addition, another difficulty for preparing antifouling polyurea is that the polymerization of isocyanate with amino is so quick (5–10 s). Such a quick reaction prevents polyurea from sufficient wetting on the substrate, resulting in poor adhesion. To slow it down, the reaction was usually carried out with a large amount of polar solvent at a low temperature.<sup>32,33</sup> On the other hand,

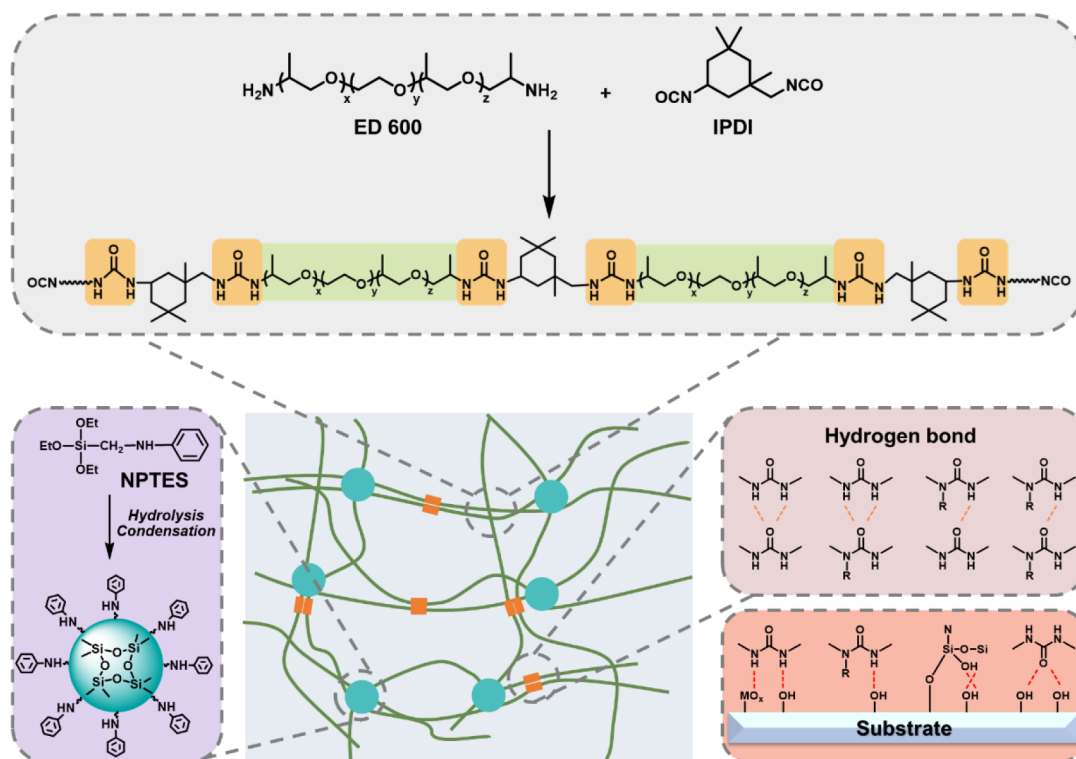
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**Figure 1.** Schematic illustration of the synthesis of the nanocluster and polyurea.

secondary amines, such as polyaspartic ester (PAE), have been used to reduce the reaction rate through steric hindrance and electron-induced effects.<sup>34–36</sup> While PAE-based polyurea can achieve adjustable gel times and good adhesion, the weaker hydrogen bonds formed by secondary amines result in relatively low tensile strength (<10 MPa).<sup>34,37</sup> To improve the tensile strength, the content of hard segments in polyurea was increased, but this usually made polyurea to be more brittle. Alternatively, nanofillers were added to improve the strength. However, phase separation led to a decrease in strength and toughness.<sup>38</sup> On the other hand, polyurea is less commonly used in biomedical engineering and marine antifouling fields since it does not have antifouling groups.

Jeffamine ED600 is an amino-terminated PPG-*b*-PEG-*b*-PPG block copolymer with a molecular weight of ~600 g/mol (Figure 1), in which the PEG content accounted for about 75%. It has good biocompatibility like PEG and can form a hydration layer with water to prevent the adhesion of protein.<sup>39–41</sup> Inspired by this, we have prepared antibiofouling polyurea with high mechanical strength and adhesion via polymerization of NCO-terminated ED600-based prepolymer with secondary amine nanoclusters. The polymerization rate can be significantly reduced so that the polyurea spreads well on the substrate. The residual silanol groups in the nanocluster and hydrogen bonding further increase the adhesion. The high cross-linking density network constructed by nanoclusters improves the mechanical strength. On the other hand, the PEG segments enable the polyurea to have antifouling ability. We investigated the wetting and adhesion of a substrate, the mechanical properties, as well as the antifouling performance of polyurea. Our aim is to develop antifouling polyurea with high adhesion, mechanical strength, and controllable operation time.

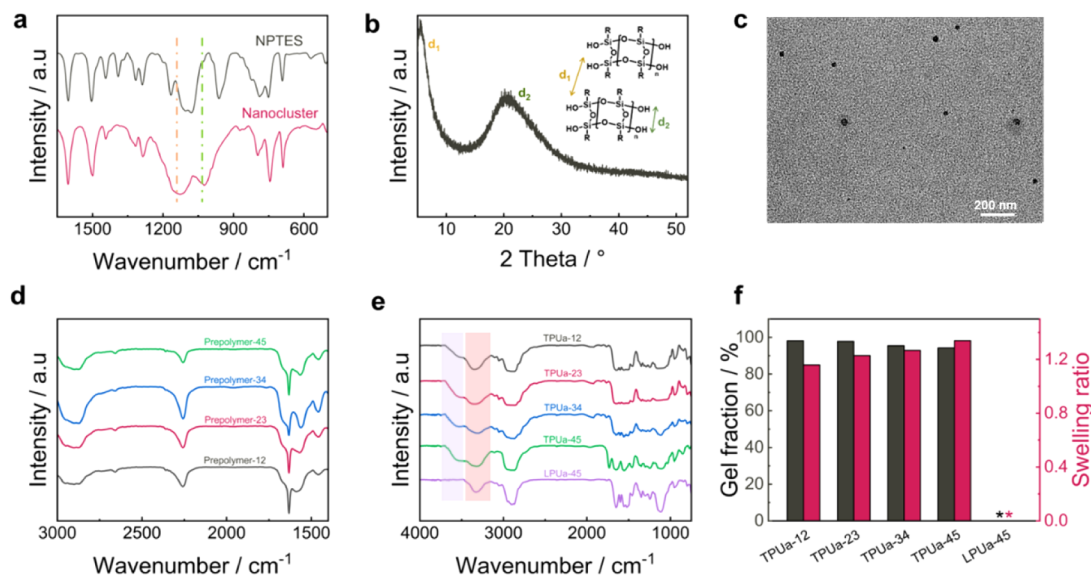
## EXPERIMENTAL SECTION

**Materials.** Isophorone diisocyanate (IPDI), *N*-phenylaminomethyltriethoxysilane (NPTES), and (3-aminopropyl)triethoxysilane were obtained from Aladdin. PPG-*b*-PEG-*b*-PPG triblock copolymer (Jeffamine ED600) was from Huntsman. The ammonia solution (NH<sub>3</sub>·H<sub>2</sub>O, 25–28 wt %), fibrinogen (*M*<sub>w</sub> = 340 kDa), and extra-dry tetrahydrofuran (THF) were from Energy Chemical. Desmophen NH1220 was purchased from Covestro Polymers. Physiological phosphate-buffered saline (PBS, pH 7.4) and deionized water were purchased from Guangzhou Yongjin Biotechnology.

**Synthesis of Nanoclusters.** First, THF (30 mL), deionized water (1.98 g), and NH<sub>3</sub>·H<sub>2</sub>O (0.57 g) were mixed and stirred for 10 min at 60 °C in a 100 mL flask. Subsequently, NPTES (20.00 g) was added to it and stirred for 8 h. THF in the resulting solution was removed by rotary evaporation and vacuum oven.

**Synthesis of NCO-Terminated Prepolymer.** ED600 was initially heated to 110 °C in a round-bottom flask under a vacuum for 2 h to remove water. Then, it was dispersed in THF. In another round-bottomed flask, IPDI and THF were stirred under N<sub>2</sub> conditions for 30 min. Subsequently, the ED600 solution was added dropwise and the mixture stirred for another 2 h to obtain the prepolymer solution. The prepolymer was denoted as prepolymer-*xy*, where *x* and *y* represent the molar ratios of ED600 and IDPI. The composition of the prepolymer is presented in Table S1.

**Preparation of Polyurea.** As illustrated in Figure 1, polyurea was prepared via polymerization of the nanoclusters with the NCO-terminated prepolymer. After the nanoclusters were dissolved in THF, the prepolymer solution was introduced and stirred for 30 min at 25 °C. It is then coated on the substrate to prepare the coating or poured into a tetrafluoroethylene mold to prepare the elastomer. Realistic photographs of the polyurea coating and elastomer are shown in Figure S1. Desmophen NH1220 is a PAE chain extender prepared by the Michael addition of 2-methylglutenediamine and diethyl maleate (Figure S2), which has been used to reduce the reactivity of polyurea.<sup>42</sup> Considering the matching of the PAE chain extender and prepolymer activities, we prepared the control sample by blending Desmophen NH1220 with prepolymer-45. The composition of the polyurea is detailed in Tables S2 and S3. According to the use of



**Figure 2.** (a) FTIR spectroscopy of NPTES and the nanocluster. (b) XRD spectroscopy of the nanocluster. (c) TEM images of the nanoclusters. (d) FTIR spectroscopy of the prepolymer. (e) FTIR spectroscopy of polyurea. (f) Gel fraction and swelling ratio of polyurea.

different prepolymers, the polyurea with a nanocluster was denoted as TPUa-xy. The control sample was named LPUa-45.

## ■ CHARACTERIZATION

**Nuclear Magnetic Resonance Spectroscopy (NMR) Measurement.** <sup>1</sup>H NMR and <sup>29</sup>Si NMR analyses were performed on a spectrometer (Bruker AV600 NMR) using CDCl<sub>3</sub> as the solvent.

**Fourier Transform Infrared (FTIR) Measurement.** The FTIR spectrum was recorded on a Nicolet IS50-Nicolet Continuum spectrometer (Thermo Fisher Scientific) with 32 scans at a resolution of 4 cm<sup>-1</sup>.

**Wide-Angle X-Ray Diffraction (XRD) Measurement.** The XRD experiment was conducted on a multipurpose high-power X-ray diffractometer (X'pert Powder, PANalytical) with CuKα radiation.

**Transmission Electron Microscope (TEM) Measurement.** The morphology of the nanocluster was characterized by TEM (JEM 2100Plus, JEOL) at 200 kV. Nanoclusters were dispersed in THF through ultrasound and dropped onto an ultrathin copper mesh to prepare samples.

**Dynamic Light Scattering (DLS) Measurement.** The nanocluster size was measured on a nanoparticle analyzer (HORIBA SZ-100Z) using the DLS method. THF was used as a solvent.

**Gel Permeation Chromatography (GPC) Measurement.** Molecular weights were measured by a 1260 Infinity (Agilent). THF was used as the eluent solvent at a flow rate of 1 mL/min at 35 °C.

**Gel Fraction (G) and Swelling Ratio (Q) Determination.** The sample (*M*<sub>0</sub>) was immersed in tetrahydrofuran (THF) for 3 d at 25 °C. Then, the sample was dried under vacuum at 80 °C for 24 h, and the weight was recorded as *M*<sub>1</sub>. The gel fraction was determined by using the following equations:

$$G = \frac{M_1}{M_0} \times 100\%$$

The sample (*M*<sub>0</sub>) was immersed in water for 3 d at 25 °C. Then, the water was removed from the sample surface, and the weight was recorded as *M*<sub>2</sub>. The swelling ratio was determined by using the following equation:

$$Q = \frac{M_2}{M_0}$$

**Tensile Test.** Tensile tests were conducted on an Instron 5966 instrument at 25 °C. The polyurea (a width of 2 mm, thickness of 0.6 mm, and length of 35 mm) underwent testing at stretching rates of 25 mm/min. A cyclic tensile test was conducted, involving stretching the sample to a 300% strain and subsequently unloading it to its original position at different rates. Another cyclic tensile test, conducted by stretching the sample to 300% strain with strain rates of 25 and 500 mm/min, was repeated for 10 cycles.

**Dynamic Mechanical Analysis.** Dynamic mechanical analysis was conducted using a dynamic mechanical analyzer (DMA) (NETZSCH DMA 242C) with a ramp rate of 10 °C/min applied from −80 to 150 °C.

**Thermal Analysis.** Thermal analysis measurements were conducted on a NETZSCH TG209F1 under N<sub>2</sub> atmosphere at a rate of 20 °C/min.

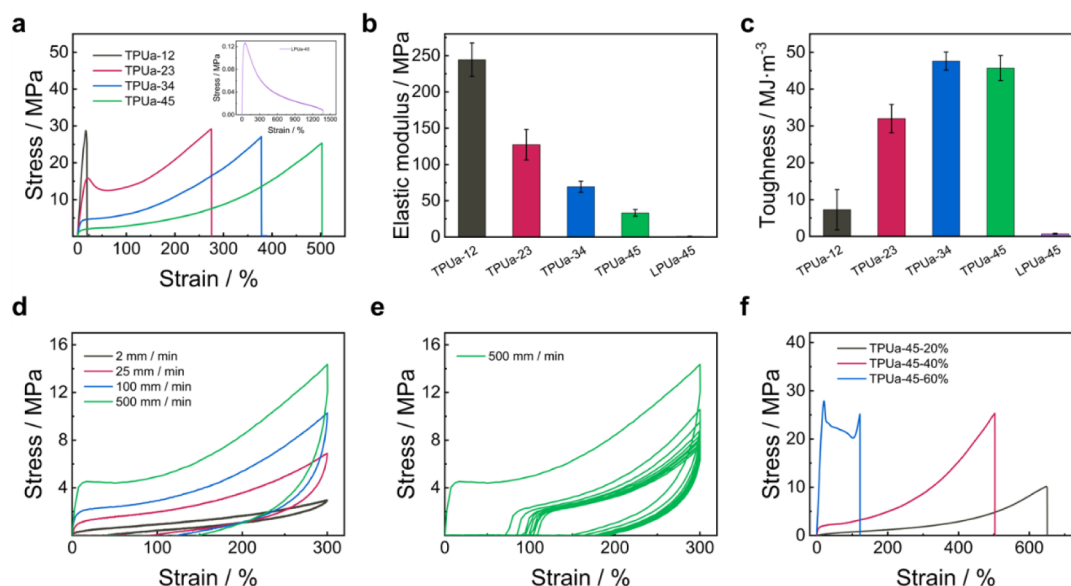
**Adhesion Measurement.** The adhesion of polyurea on a substrate was measured by an automatic tester (PosiTest, Model AT-A) with a pull rate of 0.2 MPa/s. Before coating on natural rubber, the substrate surface was treated with 80 wt % sulfuric acid to increase the roughness so that it was well wetted by polyurea.

**Energy-Dispersive X-Ray Spectroscopy (EDS) Analysis.** The EDS analysis of the polyurea was performed on a scanning electron microscope (SEM, Merlin, Zeiss) at 15 kV.

**X-Ray Photoelectron Spectroscopy (XPS).** XPS analysis was performed with an Axis Supra+ (Kratos). The reference for binding energy was set to C (1s) at 284.6 eV.

**Water Contact Angle (WCA) Measurement.** The WCA was determined utilizing a theta optical tensiometer (Attension, Biolin Scientific) by depositing a 5 μL liquid





**Figure 3.** (a) Stress–strain curves of the polyurea. (b) Elastic modulus of the polyurea. (c) Toughness of the polyurea. (d) Single-cycle curves of TPUa-45 at different rates. (e) Cycle curves of TPUa-45 at 500 mm/min. (f) Stress–strain curves of TPUa-45 with different nanocluster contents.

droplet on polyurea surface at 25 °C. The average value was computed based on measurements from 5 distinct regions.

**Quartz Crystal Microbalance with Dissipation (QCM-D).** The protein resistance of polyurea was investigated by using QCM-D (Q-sense AB, Sweden). A polyurea film was prepared by spin casting on an AT-cut quartz crystal. A blank AT-cut quartz crystal was chosen as the control sample. Fibrinogen solution (1.0 mg/mL) was transmitted to the polyurea surface at a flow rate of 150  $\mu$ L/min. Briefly, the frequency change ( $\Delta f$ ) is associated with protein adsorption, while the dissipation change ( $\Delta D$ ) correlates with the viscoelastic properties of the adsorbed protein layer. The presented data are from the third overtone.

**Antibacterial Assay.** *Staphylococcus aureus* (*S. aureus*), *Escherichia coli* (*E. coli*), and *Pseudomonas* sp. NOV776 (*P. sp.*) were selected as model bacteria. The sterilized films (1  $\times$  1 cm<sup>2</sup>) were put in a 24-well culture plate, and then 100  $\mu$ L bacterial suspension (10<sup>6</sup> cfu·mL<sup>-1</sup>) was added. After placement at 37 °C (*E. coli* and *S. aureus*) or 30 °C (*P. sp.*) for 4 h, the surface was gently rinsed three times. 100  $\mu$ L of the final washing solution by ultrasonic cleaner was dispersed in the medium and cultured at 37 °C (*E. coli* and *S. aureus*) or 30 °C (*P. sp.*) for 12 h.

**Antidiatom Test.** The diatom (*N. incerta*) was cultivated in a light incubator within a 12:12 h light-dark cycle at 23 °C. A polyurea film was fabricated on a silicon wafer with a blank silicon wafer serving as the control sample. The specimen was arranged in a 24-well microplate and submerged in a diatom suspension. After a 24-h immersion, the surface was rinsed with artificial seawater.

## RESULTS AND DISCUSSION

The molecular structure and size of the synthesized nanoclusters were first characterized. FTIR spectroscopy of the nanocluster is presented in Figure 2a. The bands at 1045 and 1150 cm<sup>-1</sup> correspond for the Si–O–Si group, indicating that the ethoxy groups in NPES have undergone a hydrolyzation-condensation reaction. <sup>29</sup>Si NMR spectroscopy shows three peaks at  $\delta$  = -52 ppm,  $\delta$  = -70 ppm, and  $\delta$  = -76 ppm

indicating the presence of residual siloxanes in the nanocluster (Figure S3). The silicon atoms in the three chemical environments showed that NPES had condensed into stereoscopic polysiloxane. The structure of the nanocluster was further characterized by XRD. As reported before, the molecular structure with a linear, double-strained siloxane backbone of stereoscopic polysiloxane exhibits two distinct diffraction peaks at 5° and 20° in XRD spectra, which are related to the intramolecular chain-to-chain distance and intrachain distance of the nanocluster, respectively.<sup>43,44</sup> As shown in Figure 2b, the XRD data also show these two characteristic peaks. This result clearly indicates that the nanoclusters have a precisely defined molecular structure with a linear, double-strained siloxane backbone (Figure S4), which is distinctly different from nanoparticles.<sup>45,46</sup> Figure 2c shows the TEM images of the nanoclusters. The nanoclusters are irregular spheres with a diameter of ~20–80 nm. This result is also supported by DLS data (Figure S5). Compared with nanoparticles in other literature, the size of nanoclusters is small enough and their distribution is narrow.<sup>47,48</sup> Therefore, it can uniformly disperse in polyurea to endow the polyurea with excellent mechanical properties without agglomeration. The successful synthesis of the prepolymer was confirmed by FTIR spectroscopy (Figure 2d) and <sup>1</sup>H NMR spectroscopy (Figure S6). The band at 2200 cm<sup>-1</sup> in Figure 2d is attributed to the NCO in the prepolymer. As the molar ratio of IPDI/ED600 increases, the length of the prepolymer chain increases (Table S4). FTIR spectra show that LPUa-45 has only one band at 3300–3500 cm<sup>-1</sup> attributed to urea units, whereas TPUa-xy also has a new band at 3300–3500 cm<sup>-1</sup> attributed to the –NH group in the nanocluster. Besides, the band at 2200 cm<sup>-1</sup> no longer appears in each TPUa-xy sample, indicating that NCO groups have reacted with secondary amines on nanoclusters and transformed into –NH groups (Figure 2e). For comparison, we used (3-aminopropyl)triethoxysilane to prepare primary amine nanoclusters<sup>49</sup> and dropped them into the NCO-terminated ED600-based prepolymer solution. However, attributed to the high reactivity of primary amine and NCO, the mixture solution gelled during the drip process

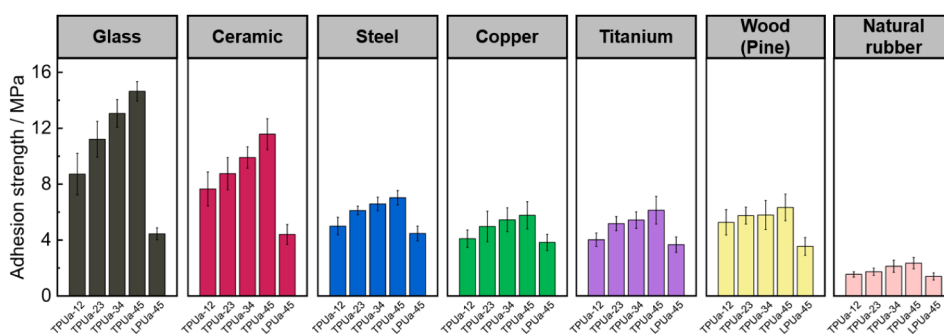


Figure 4. Adhesion of polyurea on different substrates.

(Figure S7). Figure 2f depicts the gel fraction and swelling ratio of the polyurea, reflecting the cross-linking density of polyurea. The linear LPUa-45 was completely dissolved in THF and deionized water, so the gel fraction was zero. For TPUa-*xy*, as the prepolymer chain increases, the gel fraction decreases, indicating a decrease in the cross-linking density. However, the gelation fraction of all TPUa-*xy* was higher than 90%. On the contrary, the swelling rate of the polyurea gradually increases as the prepolymer chain length increases. Yet, the swelling rate of TPUa-45 in water is very low. This is due to the hydrophobic PPG units and Si–O–Si units there.

The thermal properties of polyurea were investigated by using DMA and TGA. Figure S8 displays the  $\tan \delta$  curves. As the prepolymer chain length increases,  $\tan \delta$  gradually shifts toward lower temperature, indicating a gradual decrease in the glass transition temperature ( $T_g$ ). Since the content of ED600 is fixed, as the prepolymer chain length increases, the number of prepolymer chains decreases, leading to a decrease in cross-linking density. Due to its linear structure, LPUa-45 exhibits the lowest  $T_g$ .

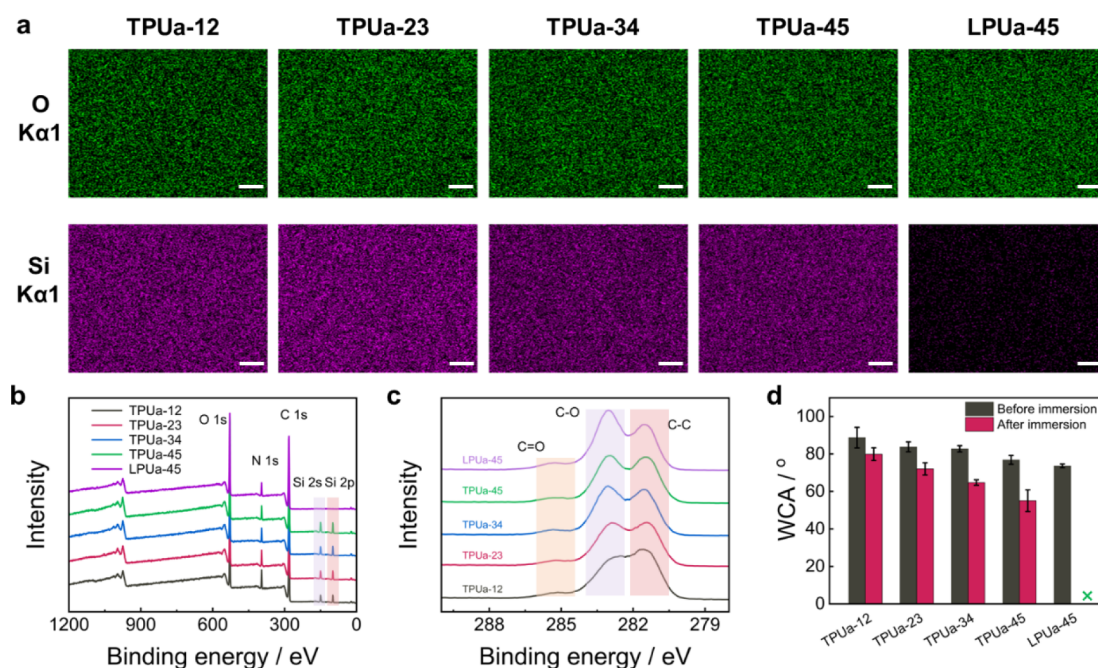
In Figure S9, each TGA curve exhibits two stages, corresponding to the degradation of the hard segments (urea units) and soft segments (ED600), respectively.<sup>50</sup> As the prepolymer chain length increases, the content of hard segments decreases; therefore, the mass loss decreases in the first stage. Namely, the thermal stability increases with the prepolymer chain length. Owing to the rigid benzene rings in the nanocluster and cross-linking network, TPUa-45 has improved thermal stability relative to LPUa-45.

Figure 3a–c presents the stress–strain curves, elastic modulus, and toughness of the polyurea. LPUa-45 exhibits a remarkably low tensile strength ( $\sim 0.13$  MPa), whereas the tensile strength of TPUa-*xy* surpasses 25 MPa, much higher than the traditional PAE-based polyurea ( $< 10$  MPa).<sup>34,37</sup> This is due to the use of secondary amines in PAE-based polyurea reducing the hydrogen bonding density while decreasing the polymerization rate of polyurea. In this study, polyfunctional secondary amine nanoclusters were synthesized and reacted with NCO-terminated prepolymers to prepare polyurea. While reducing the reaction rate of polyurea, the nanoclusters can cross-link the prepolymers into a high-density polymer network, making up for the loss of mechanical properties caused by the decrease of hydrogen bonding density. Therefore, polyurea containing secondary amine nanoclusters has both a low reaction rate and high tensile strength.

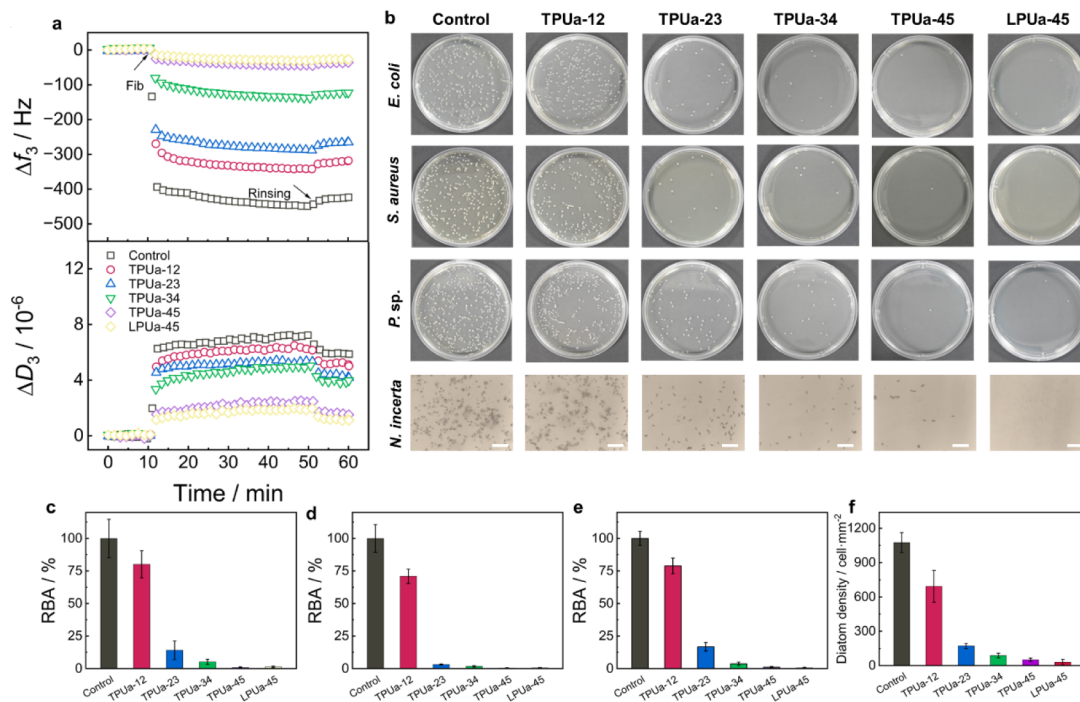
Note that as the prepolymer chain length increases, the elongation at break of TPUa-*xy* increases from  $\sim 20\%$  to  $\sim 500\%$ , whereas the tensile strength remains at about 25 MPa. The increase in mechanical strength due to more hydrogen

bonding compensates for its reduction caused by the decrease in cross-linking density. For TPUa-*xy*, the reduction in cross-linking density leads to a decrease in the elastic modulus, but it is still significantly higher than that of LPUa-45. As the prepolymer chain length increases, the toughness of TPUa-*xy* increases from  $7.27 \pm 5.46$  to  $47.61 \pm 2.47$  MJ/m<sup>3</sup>,  $\sim 10$ –66 times higher than that of LPUa-45 ( $0.72 \pm 0.15$  MJ/m<sup>3</sup>). This is because more hydrogen bonds are formed in a longer prepolymer chain with more urea units, which can dissipate more external forces. Figure 3d displays the single-cycle tensile curves of TPUa-45 as a function of the tensile rate. A higher strain rate results in a larger area enclosed by the tensile curve, indicating that more energy is consumed. This is because the polyurea contains a large number of hydrogen bonds and is cross-linked by nanoclusters; more energy is needed to overcome the restriction of hydrogen bonds and nanoclusters when subjected to external force.<sup>51</sup> The polyurea exhibits good fatigue resistance properties, and no fracture occurs after ten cycles of stretching at low (25 mm/min) and high (500 mm/min) stretching rates (Figures S10 and 3e). We also investigated the nanocluster content dependence of the mechanical properties of polyurea (Figure 3f). As the nanocluster content increases, the hard segments increase, and either its tensile strength or the elastic modulus increases. To compare the effect of different molecules on the polymerization results. We used ED900 ( $\sim 900$  g/mol) or ED2003 ( $\sim 2000$  g/mol) to replace ED600, but the tensile strength of the polyurea prepared by ED900 or ED2003 was lower than that of the polyurea prepared by ED600 (Figure S11).

Figure 4 shows the adhesion strength of polyurea on different substrates. For LPUa-45, the adhesion strength has a value on glass ( $4.45 \pm 0.42$  MPa), ceramic ( $4.42 \pm 0.71$  MPa), steel ( $4.40 \pm 0.52$  MPa), copper ( $3.83 \pm 0.58$  MPa), titanium ( $3.66 \pm 0.55$  MPa), wood ( $3.54 \pm 0.63$  MPa), or natural rubber ( $1.39 \pm 0.26$  MPa). Compared with LPUa-45, the adhesion of TPUa-*xy* is significantly improved, indicating that the introduction of nanoclusters improves the strength and cohesion energy of the polyurea. Besides, as the prepolymer chain length increases, the adhesion of TPUa-*xy* increases. TPUa-45 has the highest adhesion strength on either glass ( $14.64 \pm 0.69$  MPa), ceramic ( $11.57 \pm 1.11$  MPa), steel ( $7.03 \pm 0.51$  MPa), copper ( $5.78 \pm 0.97$  MPa), titanium ( $6.13 \pm 0.99$  MPa), wood ( $6.33 \pm 0.95$  MPa) or natural rubber ( $2.35 \pm 0.41$  MPa). This is because the number of urea units in the longer prepolymer increases, forming more hydrogen bonds with the substrate. In addition, the adhesion of TPUa-45 on different substrates is also different. In the case of ceramic or glass, besides the hydrogen bonding, the silanol groups



**Figure 5.** Characterization of the polyurea surface: (a) EDS images, (b) XPS spectra, (c) High-resolution XPS spectra, and (d) WCA. The scale bar is 5  $\mu\text{m}$ .



**Figure 6.** (a) QCM-D curves on the surfaces of polyurea. (b) Colonial morphology images of *E. coli*, *S. aureus*, *P. sp.*, and *N. incerta* on the surfaces of polyurea. (c–e) RBA of *E. coli*, *S. aureus*, and *P. sp.* (f) Diatom density of *N. incerta*. The scale bar is 50  $\mu\text{m}$ .

hydrolyzed from alkoxy groups in the nanocluster can react with the hydroxyl groups on the substrate; that is, polyurea chemically bonds to the substrate; therefore, the adhesion is the highest.<sup>52</sup> The high adhesion on steel, copper, or titanium arises from the hydrogen bonding between the polyurea and the substrate. The silanol groups hydrolyzed in the nanoclusters cannot chemically bond to the substrate, so the adhesion on steel, copper, or titanium is lower than on glass or ceramics. For wood with a porous structure, slow polymer-

ization allows the prepolymer and nanocluster to penetrate the pores, so adhesion is high. However, due to the low strength of wood, it is prone to breakage during testing, resulting in the separation of polyurea from the substrate. Therefore, the adhesion of TPUa-45 on wood is slightly lower than that on steel, copper, or titanium. For the natural rubber, its surface has a low surface energy and low polarity. There is no hydrogen bonding or chemical bonding interaction between



TPUa-45 and the rubber surface, so the adhesion of TPUa-45 on rubber is the lowest.

The surface properties of the polyurea were studied by using EDS, XPS, and WCA measurements. Figure 5a presents the EDS images. Both oxygen and silicon elements show uniform distribution on TPUa-xy, indicating that the nanoclusters are uniformly distributed in polyurea without agglomeration. This is because nanoclusters with hyperbranched structures can be well dispersed in THF, making them easy to process into uniform films with low roughness.<sup>53,54</sup> Due to the absence of nanoclusters, silicon element was not detected on the LPUa-45 surface. Figure 5b,c shows the XPS spectrogram. The peaks at 150 and 100 eV correspond to Si (2s and 2p) in the nanocluster, while the peaks at 285 and 528 eV correspond to C 1s and O 1s in the nanocluster and prepolymer. The absence of Si (2s and 2p) peaks in LPUa-45 indicates no nanocluster. Despite a marginal difference in element content (Figure S12), high-resolution C 1s scans reveal a new peak attributed to C–O from ED600 (shifted 1.5 eV from C–C peaks) (Figure 5c).<sup>55</sup> Furthermore, the intensity of the C–O peak from ED600 increases with prepolymer chain length, indicating more ED600 on the surface.

To examine the effect of chain length on the hydrophilicity of polyurea, we measured WCA on the polyurea surface. As shown in Figure 5d, before soaking in water, the WCA of TPUa-12 was about 90°, indicating a low hydrophilicity surface. This is because the prepolymer chains are too short to be on the surface, and most of the surface is covered by Si–O–Si units. As the prepolymer chain lengthens, the WCA decreases significantly, indicating a more hydrophilic surface. The change in surface hydrophilicity is due to the difference in the ED600 content on the surface of these samples. As indicated in Tables S1 and S4, although the amounts of ED600 are the same, the prepolymer chain length is different. As the prepolymer chain length increases, the chain flexibility of TPUa-xy increases, making a more hydrophilic ED600 enrichment onto the surface. The fact is supported by the high-resolution XPS spectra (Figure 5c). So, the surface hydrophilicity increases. After a 96-h soaking in deionized water, the WCAs of each sample significantly decreased. This is attributed to the hydrophilic PEG segments and hydrophobic PPG segments in ED600. Upon exposure to air, the PPG segments influence the surface properties. After immersion, the hydrophilic PEG segments appear on the surface, increasing the surface hydrophilicity.

The protein resistance measurement was conducted using a QCM-D with fibrinogen as the model protein. Figure 6a illustrates the time dependence of the changes in frequency ( $\Delta f$ ) and dissipation ( $\Delta D$ ) concerning protein adsorption. For uncoated quartz crystals, notable changes in  $\Delta f$  and  $\Delta D$  indicate significant protein adsorption. For TPUa-12, there is a reduction in  $-\Delta f$ . Furthermore, as the chain length of the prepolymer increases,  $-\Delta f$  gradually decreases, indicating that protein absorption decreases or protein resistance increases. In the case of TPUa-45 or LPUa-45, either  $\Delta f$  or  $\Delta D$  is close to zero, indicating their good protein resistance. This is due to the increase in the prepolymer chain, and more PEG segments can enrich onto the surface. The enriched PEG can form a hydration layer with interfacial water to effectively resist protein adsorption.<sup>56–58</sup>

Antibacterial assays (*E. coli*, *S. aureus*, and *P. sp.*) and antidiatoms (*N. incerta*) test were conducted to assess the antifouling properties of polyurea. It is worth noting that *P. sp.*

was extracted from the panels immersed in field tests and *N. incerta* is derived from natural seawater. They are widely used models to reflect the marine antifouling performance and guide to the real applications.<sup>59–61</sup> Figure 6b shows the optical photographs of agar plates dispersed by suspension of *E. coli*, *S. aureus*, and *P. sp.* and the microscopic images of the diatom (*N. incerta*). Figure 6c–e shows their relative bacterial adsorption (RBA). There is a large number of bacteria on surfaces of the control sample and TPUa-12, indicating poor bacterial resistance. Compared with TPUa-12, the number of bacteria on TPUa-23 decreased significantly. This is because the PEG content on the TPUa-23 surface is significantly higher than that on the TPUa-12 surface, so TPUa-23 can inhibit the adsorption of bacteria more effectively. As the prepolymer chain length increases, the observable bacteria on polyurea decrease significantly. A few bacteria were observed on the TPUa-45 and LPUa-45 surface (RBA < 10%). This is because these two samples have the highest PEG content on the surface, so they had the best resistance to adsorption of bacteria. Considering the mechanical properties and antibacterial properties, TPUa-45 was selected as a representative sample for the long-term stability test. First, the TPUa-45 coating was soaked in water for a week. After immersion, TPUa-45 did not peel off from the substrate. More importantly, the antibacterial properties of the soaked coatings were also tested. As shown in Figure S13, TPUa-45 still had good antibacterial activity against the three bacteria. This result indicates that the antifouling coating has good long-term stability.

The antidiatom experiment showed a similar trend as the antibacterial experiment. A high density of diatoms was observed on the surface of the control sample (1075 cells/mm<sup>2</sup>) or TPUa-12 (692 cells/mm<sup>2</sup>) (Figure 6b). As the prepolymer chain length increased, the diatom density decreased from 692 cells/mm<sup>2</sup> to 51 cells/mm<sup>2</sup>. Namely, polyurea exhibits good antidiatom performance when the prepolymer is long enough. In summary, polyurea has excellent fouling resistance against proteins, bacteria, as well as diatoms. Actually, it is well-known that PEG can resist the adhesion of extracellular polymeric substances (EPS) or biofilm by forming a hydration layer.<sup>62,63</sup> Considering that polysaccharide is also a major component of EPS, we think that these materials could be able to resist the adsorption of polysaccharide since they contain a high content of PEG segments.

## CONCLUSIONS

We have synthesized nanoclusters via the hydrolysis of *N*-phenylaminomethyltriethoxysilane. The polymerization of such nanoclusters with NCO-terminated prepolymers consisting of PPG-*b*-PEG-*b*-PPG segments yields a high-performance polyurea. The nanoclusters contain low-reactivity secondary amines and can slow down the polymerization so that the wetting and adhesion of the polyurea on a substrate are improved. The residual silanol groups on the nanoclusters further increased the adhesion. Polyurea has a suitable operation time, high adhesion, and robust mechanical properties. Particularly, it exhibits excellent antifouling performance due to the PEG segments. Compared with traditional antifouling polyurea, we have solved the problems of too fast polymerization, agglomeration of nanofiller, poor static antifouling ability, and low mechanical strength by using polyfunctional secondary amine nanoclusters and NCO-terminated ED600-based prepolymers. The polyurea is

expected to find applications in biomedical engineering and marine antifouling. This work provides a new strategy to develop high-performance functional polymers.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.4c02070>.

The photographs of the polyurea coating and elastomer; the molecular structure of Desmophen NH1220 and nanocluster;  $^{29}\text{Si}$  NMR, DLS curve of nanocluster;  $^1\text{H}$  NMR and molecular weight of prepolymer; photograph of a gel of primary amine nanoclusters reacting with prepolymer-45 and solution of secondary amine nanoclusters reacting with prepolymer-45. TGA curves of polyurea; cycle curves of TPUa-45 at 25 mm/min; stress-strain curves of polyurea containing ED600, ED900, or ED2003; adhesion of the polyurea on polyethylene glycol terephthalate; oxygen content on polyurea, long-term stability testing of TPUa-45; compositions of prepolymer and polyurea; compositions of prepolymer, polyurea, and TPUa-45 with different nanocluster content (PDF)

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### Notes

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